THE ALGEBRAIZED MAP IS PROPER EXACTLY WHEN THE FORMAL MAP IS A FORMAL MODIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete: both directions are given and the three DVR cases are each carried out

Editorial note. The label lemma-output-proper names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the dilatation correspondence of the chapter between maps f : X' to X and rig-etale maps g : W to X_{/T}
- formal modifications
- the DVR lifting criterion for properness of morphisms of algebraic spaces and its rig-surjective variant
- the quasi-compactness and separatedness output lemmas for the correspondence
- algebraization of adic morphisms of formal spectra into algebraic spaces

Source reference. Original reference 0GDK: sources/stacks-restricted/pinned-original-source.tex, source lines 6275–6275. Prerequisite pointer invoked at proof line(s) 7283; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0AQ3: sources/stacks-chow/tags.txt, source lines 13496–13496. Prerequisite pointer invoked at proof line(s) 7326; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0APV: sources/stacks-chow/tags.txt, source lines 13488–13488. Prerequisite pointer invoked at proof line(s) 7325; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AQH: sources/stacks-chow/tags.txt, source lines 13510–13510. Prerequisite pointer invoked at proof line(s) 7332; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDM: sources/stacks-restricted/pinned-original-source.tex, source lines 6346–6346. Prerequisite pointer invoked at proof line(s)

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7288; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ARU: sources/stacks-restricted/pinned-original-source.tex, source lines 7117–7117. Prerequisite pointer invoked at proof line(s) 7290; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ARW: sources/stacks-restricted/pinned-original-source.tex, source lines 7198–7198. Prerequisite pointer invoked at proof line(s) 7290; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0ARK: sources/stacks-chow/tags.txt, source lines 13548–13548. Prerequisite pointer invoked at proof line(s) 7294; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0ARL: sources/stacks-chow/tags.txt, source lines 13549–13549. Prerequisite pointer invoked at proof line(s) 7295; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ARB: sources/stacks-restricted/pinned-original-source.tex, source lines 7015–7015. Prerequisite pointer invoked at proof line(s) 7280; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With assumptions and notation as in Theorem 0ARB let $f : X' \to X$ correspond to $g : W \to X_{/T}$. Then f is proper if and only if g is a formal modification (Definition 0GDK).*

Proof. If f is proper, then g is a formal modification by Lemma 0GDM. Assume g is a formal modification. By Lemmas 0ARU and 0ARW we see that f is quasi-compact and separated.

By Cohomology of Spaces, Lemma 0ARK and Remark 0ARL it suffices to show that given any commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X' \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(R) & \xrightarrow{p} & X \end{array}$$

where R is a complete discrete valuation ring with fraction field K, there is a dotted arrow making the diagram commute. There are three cases: Case I: $p(\mathrm{Spec}(R)) \subset U$. This case is trivial because $U' \to U$ is an isomorphism. Case II: p maps $\mathrm{Spec}(R)$ into T. This case follows from our assumption that $g : W \to X_{/T}$ is proper. Namely, if Z denotes the reduced induced closed subspace structure on T, then p factors through Z and

$$W \times_{X_{/T}} Z = X' \times_X Z \longrightarrow Z$$

is proper by assumption which implies we get the lifting property by Cohomology of Spaces, Lemma 0ARK applied to the displayed arrow. Case III: $p(\mathrm{Spec}(K))$ is not

in T but p maps the closed point of $\mathrm{Spec}(R)$ into T. In this case the corresponding morphism

$$p_{/T} : \mathrm{Spf}(R) \longrightarrow X'_{/T} = W$$

is an adic morphism (by Formal Spaces, Lemma 0APV and Definition 0AQ3). Hence our assumption that $g : W \to X_{/T}$ be rig-surjective implies we can lift $g_{/T}$ to a morphism $\mathrm{Spf}(R') \to W = X'_{/T}$ for some extension of complete discrete valuation rings $R \subset R'$. Algebraizing the composition $\mathrm{Spf}(R') \to X'$ using Formal Spaces, Lemma 0AQH we find a morphism $\mathrm{Spec}(R') \to X'$ lifting p as desired. □